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"Truth is indestructible, virtue is indestructible, purity is indestructible." – Swami Vivekananda

Introduction to Data Analytics for IoT



- One of the biggest challenges in IoT:
 - **Management** of massive amounts of **data** generated by sensors.
- Few great examples
 - commercial aviation industry
 - utility industry
- Modern jet engines are fitted with **thousands of sensors** that generate a whopping **10GB of data per second**
- A **twin engine** commercial aircraft with these engines operating on **average 8 hours a day** will generate over **500 TB** of data daily, and this is just the data from the engines!



Commercial Jet Engine

Structured v/s Unstructured Data

- Not all data is the same
- it can be categorized and thus analyzed in different ways.

- **Structured data :**

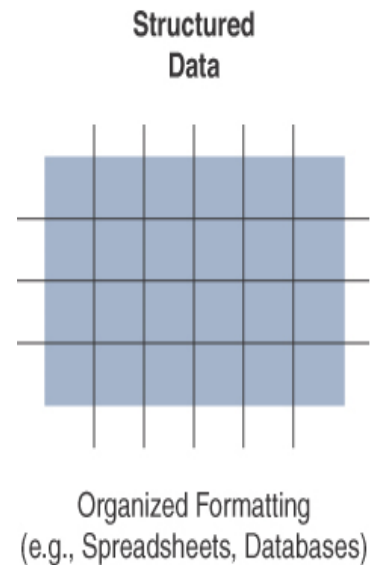
- data follows a model/schema
- defines data representation
- **e.g.** Relational Database
- easily formatted, stored, queried, and processed
 - has been core type of data used for making business decisions
 - Wide array of data analytics tools are available

- **Unstructured data:**

- lacks a logical schema
- Doesn't fit into predefined data model
- **e.g.** text, speech, images, video

- **Semi-structured data:**

- hybrid of structured and unstructured data
- Not relational, but contains a certain schema
- **e.g.** Email : fields are well defined, but body and attachments are unstructured

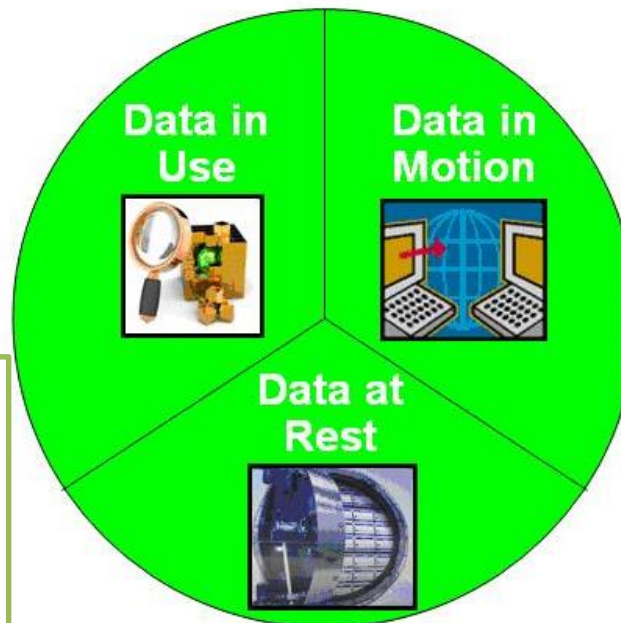


Data in Motion v/s at Rest v/s in Use

- Different states of digital data can be
 - in **transit** (data in motion)
 - being **held/stored** (data at rest)
 - being **processed** (data in use)

- **Data in motion** is data that is currently travelling across a network or
- sitting in a computer's RAM ready to be read, updated, or processed.

Data in Use:
Active data under constant change stored physically in databases, data warehouses, spreadsheets etc.



Data in Motion:
Data that is traversing a network or temporarily residing in computer memory to be read or updated.

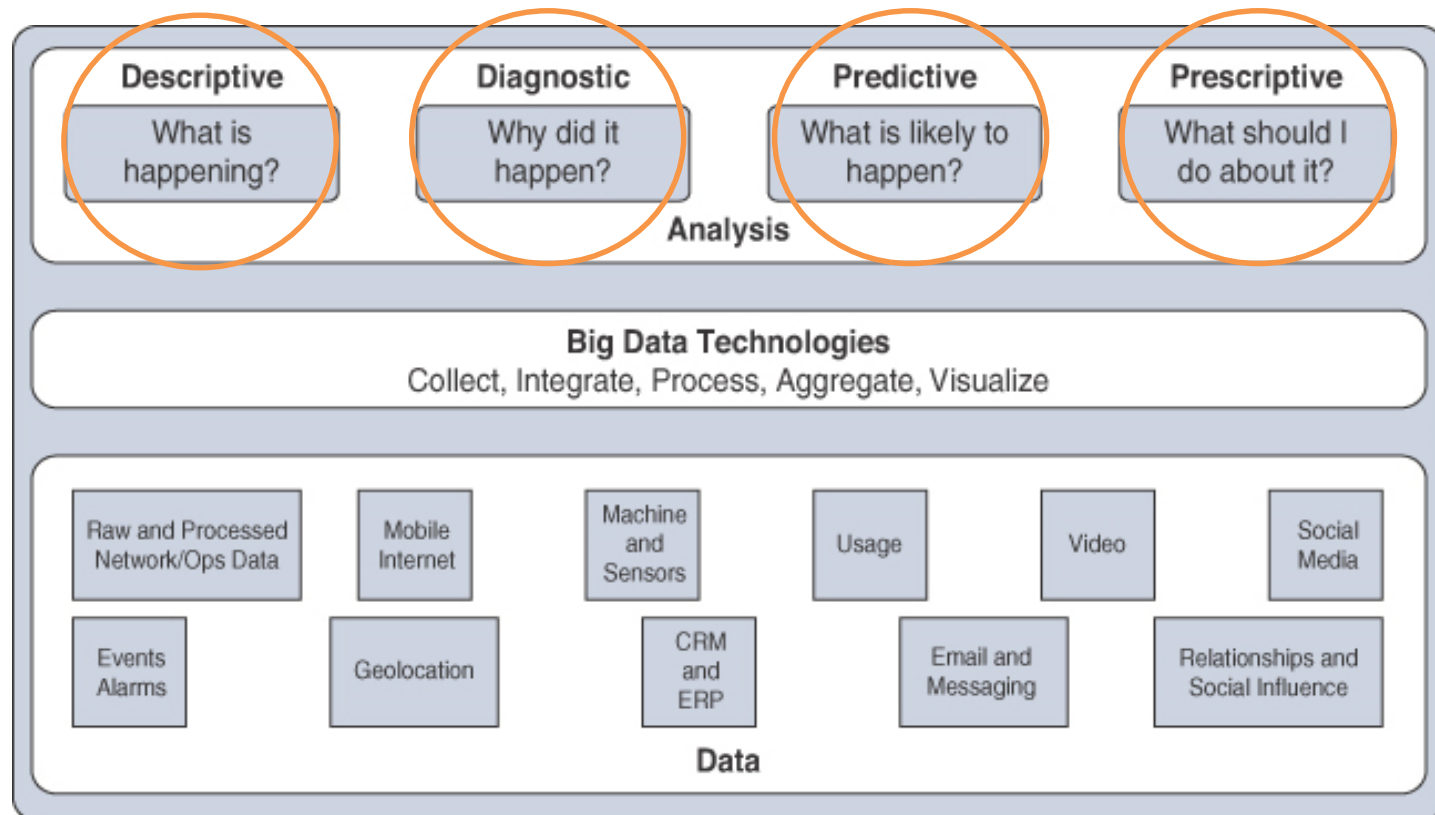
Data at Rest:
Inactive data stored physically in databases, data warehouses, spreadsheets, archives, tapes, off-site backups etc.

- **Data being processed** by one/more applications.
- data in the process of being generated, viewed, updated, appended, or erased.

- **Data at rest** is typically in a stable state.
- It is not travelling within the system or network, and
- it is not being acted upon by any application or the CPU.

IoT Data Analytics Overview

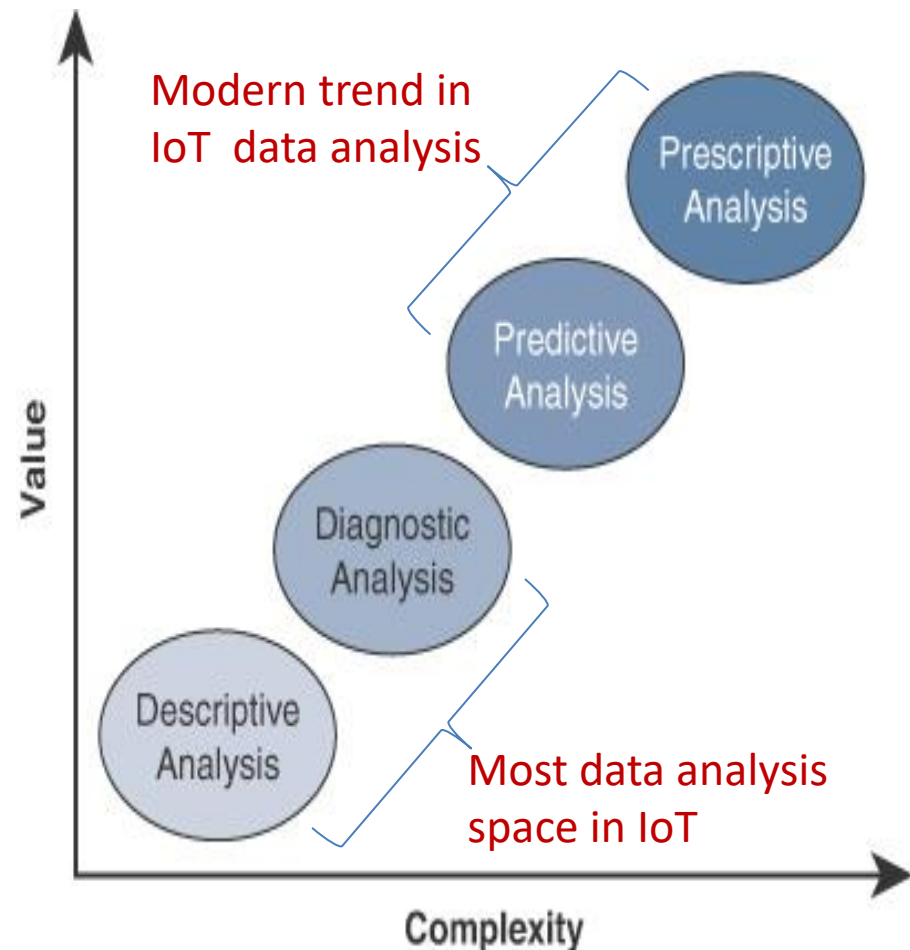
- The true **importance of IoT data** from smart objects is realized only when
 - the analysis of the data leads to **actionable business intelligence** and **insights**.
- Data analysis is typically broken down by
 - the **types of results** that are produced.



Cont...

- **Descriptive**
 - It tells you what is happening, either now or in the past.
 - e.g., thermometer in a truck engine reports temperature values every second.
- **Diagnostic**
 - It can provide the answer to “why” it has happened
 - e.g. why the truck engine failed
- **Predictive**
 - It aims to foretell problems or issues before they occur.
 - e.g., it could provide an estimate on the remaining life of the truck engine.
- **Prescriptive**
 - It goes a step beyond predictive and recommends solutions for upcoming problems.
 - e.g. it might calculate various alternatives to cost-effectively maintain our truck.

Application of Value and Complexity Factors to the Types of Data Analysis



IoT Data Analytics Challenges



- Traditional data analytics solutions **are not always adequate**
 - It typically employs a standard RDBMS and corresponding tools
- IoT data places two specific **challenges on relational database data**:
 - **Scaling problems**:
 - large number of smart objects continually send data,
 - relational databases **grow incredibly large very quickly**.
 - Results in performance issues which is costly to resolve
 - **Volatility of data**:
 - In RDBMS, **schema** is designed from the beginning,
 - changing the scheme later creates problem.
 - IoT data is **volatile** in the sense that the data model is likely to change and evolve over time.
 - A **dynamic schema** is often required.
- **Solution**: NoSQL database is used
 - does not use SQL to interact with the database
 - do not enforce a strict schema
 - support a complex, evolving data model
 - databases are inherently much more scalable

Cont...



- IoT brings **challenges to streaming and network analytics**
 - with the **live streaming** nature of its data, and
 - with **managing data** at the network level.
 - usually of a very high volume
 - real-time analysis of streaming data
 - Google, Microsoft, IBM, etc., have **streaming analytics** offerings
 - with the area of network data i.e. **network analytics**.
 - it can be challenging to ensure that the data flows are effectively managed, monitored, and secure.
 - **Network analytics tools**: Flexible NetFlow, IPFIX

Machine Learning



- How to make sense of the data that is generated in IoT?
 - by **Machine Learning**
 - ML is used to **find the data relationships** that will lead to **new business insights**
- **ML** is a part of a larger set of technologies commonly grouped under the term **artificial intelligence (AI)**.
- **AI** includes any technology that allows a computing system to mimic human intelligence
 - e.g., an App that can help you find your parked car.
 - e.g., a GPS reading of your position at regular intervals calculates your speed.
- In more **complex cases**, **static rules** cannot be simply inserted into the program
 - because the programs require parameters that can change.
 - e.g., **dictation program** – it does not know your accent, tone, speed, and so on. You need to record a set of predetermined sentences to help the tool. This process is called machine learning.

Types of ML



- **Supervised:** All data is labeled and the algorithms learn to predict the output from the input data.
- **Unsupervised:** All data is unlabeled and the algorithms learn to inherent structure from the input data.

Supervised	Unsupervised	Semi-Supervised	Reinforcement
• Data has known labels or output	• Labels or output unknown • Focus on finding patterns and gaining insight from the data	• Labels or output known for a subset of data • A blend of supervised and unsupervised learning	• Focus on making decisions based on previous experience • Policy-making with feedback
• Insurance underwriting • Fraud detection	• Customer clustering • Association rule mining	• Medical predictions (where tests and expert diagnoses are expensive, and only part of the population receives them)	• Game AI • Complex decision problems • Reward systems

Few ML Algorithms

Machine Learning Algorithms *(sample)*

	<u>Unsupervised</u>	<u>Supervised</u>
<u>Continuous</u>	• Clustering & Dimensionality Reduction ○ SVD ○ PCA ○ K-means	• Regression ○ Linear ○ Polynomial • Decision Trees • Random Forests
<u>Categorical</u>	• Association Analysis ○ Apriori ○ FP-Growth • Hidden Markov Model	• Classification ○ KNN ○ Trees ○ Logistic Regression ○ Naive-Bayes ○ SVM

Examples from IoT



Supervised Learning

- Suppose you are training a system to recognize when there is a human in a mine tunnel.
 - sensor equipped with a basic camera can capture shapes
 - return them to a computing system.
 - hundreds or thousands of images are fed into the machine.
 - each image is labelled (human or nonhuman in this case)
 - An algorithm is used to determine common parameters and common differences between the images.
 - This process is called *training*.
 - Each new image is compared with “good images” of human as per training model
 - This process is called *classification*.
 - the machine should be able to recognize human shapes.
 - the learning process is not about classifying in two or more categories but about finding a correct value.
 - *regression* predicts numeric values, whereas *classification* predicts categories.

Unsupervised Learning

- Suppose that you are processing IoT data from a factory manufacturing small engines. You know that about **0.1% of the produced engines on average need adjustments** to prevent later defects.
- Your task is to **identify them** before they shipped away from the factory.
 - you can test each engine
 - record multiple parameters, such as sound, pressure, temperature of key parts, and so on.
 - Once data is recorded, you can graph these elements in relation to one another.
 - You can then input this data into a computer and use mathematical functions to find groups.
 - A standard function to operate this grouping, *K-means clustering*
 - Grouping the engines this way can quickly reveal several types of engines that all belong to the same category.
 - There will occasionally be an engine in the group that **displays unusual characteristics**
 - This is the engine that you send for manual evaluation
 - this determination is called *unsupervised learning*.

Applications of ML for IoT



It revolves around four major domains:

I. Monitoring

- ML can be used with monitoring to detect early failure conditions or to better evaluate the environment

II. Behavior control

- Monitoring commonly works in conjunction with behavior control.
- When a given set of parameters reach a target threshold, monitoring functions generate an alarm OR would trigger a corrective action

III. Operations optimization

- The objective is not merely to pilot the operations but to improve the efficiency and the result of these operations.
- e.g., NN based system for a water purification plant in a smart city estimate the best chemical and stirring mix for a target air temperature

IV. Self-healing, self-optimizing

- The system becomes self-learning and self-optimizing.
- ML engine can be programmed to dynamically monitor and combine new parameters, and automatically deduce and implement new optimizations

What is Big Data?

- Industry looks to **three V's** to categorize big data

Velocity refers to **how quickly** data is being collected and analysed.

Hadoop Distributed File System is designed to process data very quickly.

VOLUME

- ◆ Amount of data generated
- ◆ Online & offline transactions
- ◆ In kilobytes or terabytes
- ◆ Saved in records, tables, files



Volume refers to the **scale** of the data.

It is common to see clusters of servers for storing and processing data

VELOCITY

- ◆ Speed of generating data
- ◆ Generated in real-time
- ◆ Online and offline data
- ◆ In Streams, batch or bits



VARIETY

- ◆ Structured & unstructured
- ◆ Online images & videos
- ◆ Human generated - texts
- ◆ Machine generated - readings



Variety refers to **different types** of data.

Hadoop is able to collect and store all three types – structured, unstructured, or semi-structured.

Characteristics of Big Data



- can be defined by the sources and types of data
 - Machine data
 - generated by IoT devices and is typically unstructured data.
 - Transactional data
 - from the sources that produce data from transactions on these systems, and, have high volume and structured.
 - Social data
 - which are typically high volume and structured.
 - Enterprise data
 - data that is lower in volume and very structured.

Database Technologies



- Relational databases and historians are mature technologies
 - **Relational databases**, such as Oracle and Microsoft SQL, are good for transactional, or process, data.
 - **Historians** are optimized for time-series data from systems and processes
- Database technologies used in an IoT context.
 - **NoSQL**
 - It is not a specific database technology; rather, it is an umbrella term that encompasses several different types of databases.
 - Can **quickly ingest** rapidly changing data
 - Can be **able to query** and analyse data within the database itself
 - built to **scale horizontally** i.e. database can span to multiple hosts (so distributed)
 - Best fit for IoT data:
 - **Document stores**: stores semi-structured data, such as **XML** or **JSON**.
 - » **allowing the database schema to change quickly**
 - **Key-value stores**: stores associative arrays where a key is paired with a value.
 - » capable of handling **indexing** and **persistence**.
 - **Massively Parallel Processing**
 - built on the concept of the **relational data warehouses**
 - designed to allow for **fast query processing**; often have **built-in analytic functions**
 - designed in a **scale-out architecture** such that both data and processing are distributed across multiple systems
 - **Hadoop**

Hadoop

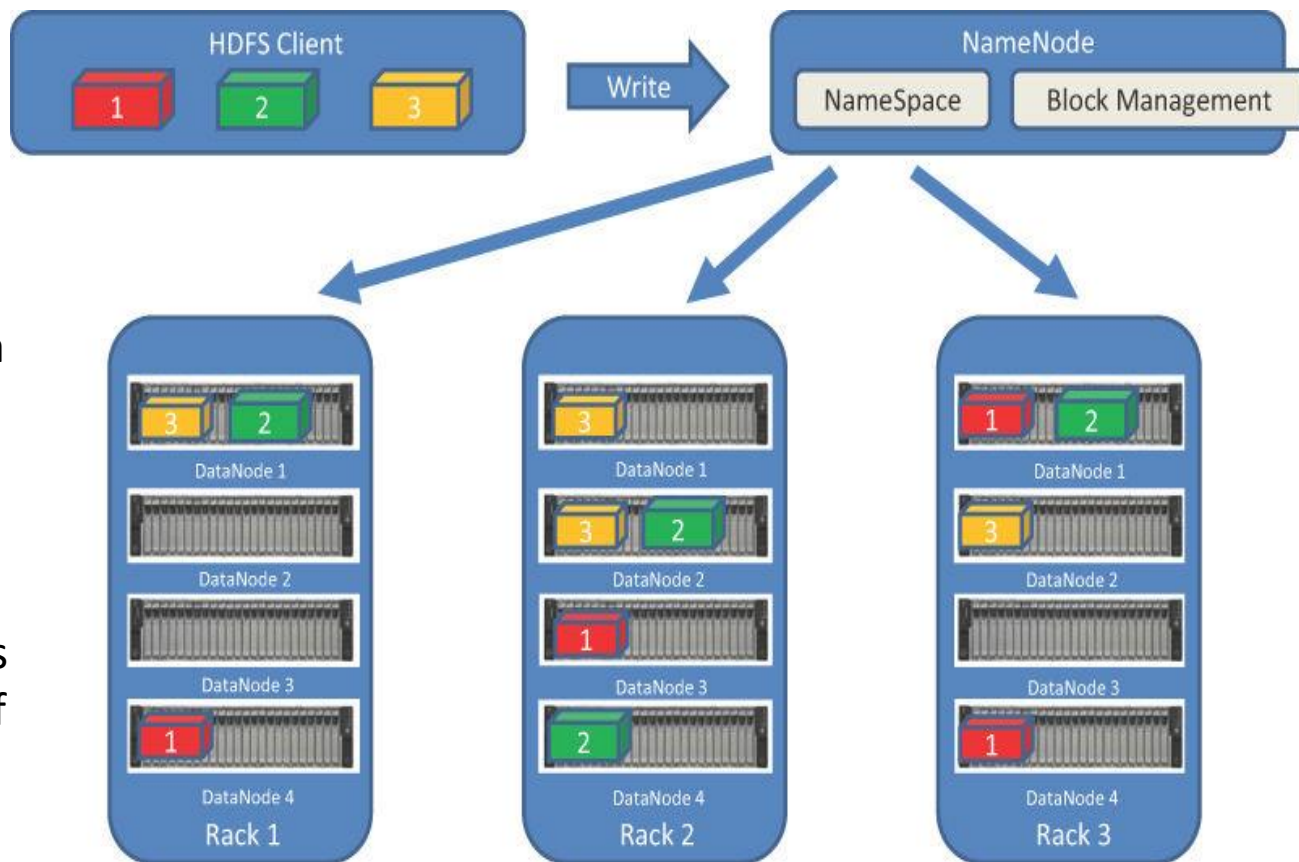


- Most popular choice as a **data repository** and **processing engine**
- Originally developed as a result of projects at Google and Yahoo!
 - original intent was to **index millions of websites** and **quickly return search results** for open source search engines.
- Initially, the project had **two key elements**:
 - **Hadoop Distributed File System (HDFS)**: A system for storing data across multiple nodes
 - **MapReduce**: A distributed processing engine that splits a large task into smaller ones that can be run in parallel
- Hadoop relies on a scale-out architecture
- Both MapReduce and HDFS
 - take advantage of this distributed architecture to store and process massive amounts of data
 - leverages local processing, memory, and storage from all nodes in the cluster

Cont...

- For HDFS, this capability is handled by specialized nodes in the cluster – NameNode and DataNode

- NameNode** coordinate where the data is stored, and maintain a map of where each block of data is stored and where it is replicated.
- DataNodes** are the servers where the data is stored at the direction of the NameNode.



Hadoop Ecosystem



- Hadoop plays an increasingly big role in the collection, storage, and processing of IoT data
- Hadoop Ecosystem **comprises of more than 100 software projects** under the Hadoop umbrella
 - Capable of every element in the data lifecycle, from data collection, to storage, to processing, to analysis and to visualization
- Several of these packages
 - **Apache Kafka**
 - **Apache Spark**
 - **Apache Storm**
 - **Apache Flink**
 - **Lambda Architecture**

Edge Streaming Analytics

- In the world of IoT vast quantities of data are generated on the fly
 - Often they are time sensitive i.e. needs immediate attention,
 - waiting for deep analysis in the cloud simply isn't possible.
 - e.g., **automobile racing industry**
 - Formula One racing car has 150-200 sensors that generate more than 1000 data points per second
 - enormous insights leading to better race results can be gained **by analyzing data on the fly**
- Big Data tools like Hadoop and MapReduce are not suited for real-time analysis
 - because of distance from the IoT endpoints and the network bandwidth requirement
- **Streaming analytics** allows you to **continually monitor and assess data in real-time** so that you can adjust or fine-tune your predictions as the race progresses.
- In IoT, streaming analytics is **performed at the edge** (either at the sensors themselves or very close to them such as gateway)
- The edge isn't in just one place. The edge is highly distributed.
- **Does the streaming analytics replaces big data analytics in the cloud?**
 - Not at all.
 - **Big data** analytics is focused on large quantities of **data at rest**, **edge analytics** continually processes streaming flows of **data in motion**.

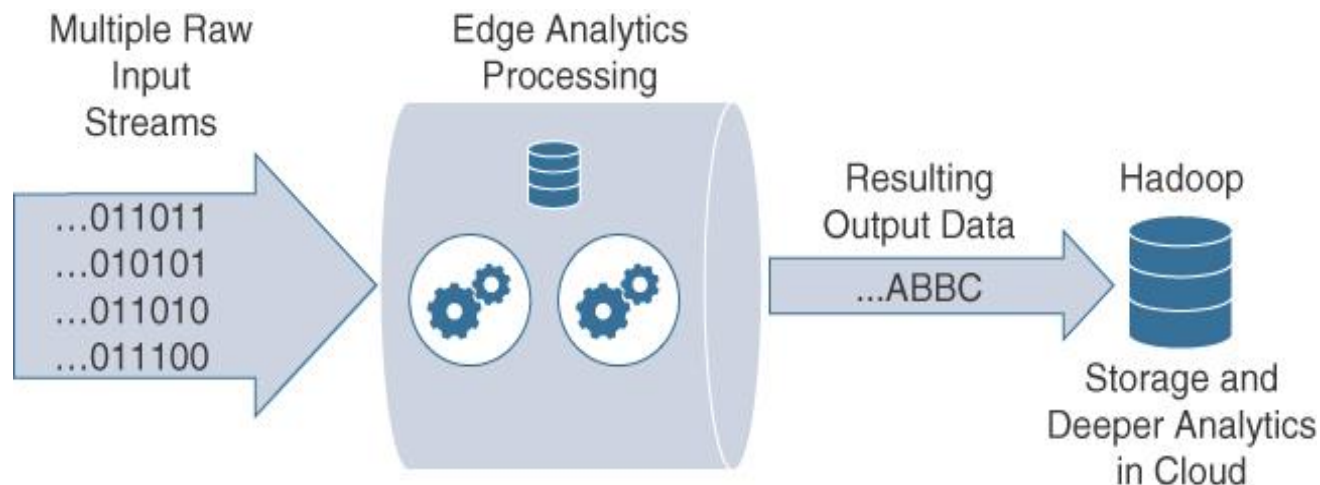
Key Values of Edge Streaming Analytics



- Reducing data at the edge
 - Passing all data to the cloud is inefficient and is unnecessarily expensive in terms of bandwidth and network infrastructure.
- Analysis and response at the edge
 - Some data is useful only at the edge and for small window of time
 - e.g., Roadway sensors combined with GPS wayfinding apps may tell a driver to avoid a certain highway due to traffic. This data is valuable for only a small window of time.
- Time sensitivity
 - When timely response to data is required, passing data to the cloud for future processing results in unacceptable latency.

Edge Analytics Core Functions

- **Raw input data**
 - This is the raw data coming from the sensors into the analytics processing unit.
- **Analytics processing unit (APU)**
 - The APU filters and combines (or separates) data streams, organizes them by time windows, and performs various analytical functions.
- **Output streams**
 - The data that is output is organized into insightful streams and passed on for storage and further processing in the cloud.



Network Analytics



- This form of analytics extremely important in managing IoT systems
- **Data analytics**: concerned with finding patterns in the data generated by endpoints
- **Network analytics**: concerned with discovering patterns in the communication flows
 - It is network-based analytics
 - power to analyze details of communications patterns made by protocols
 - correlate this pattern across the network
 - allows to understand what should be considered normal behavior in a network

Benefits



Benefits of Network Analytics:

- Offer capabilities to cope with capacity planning for scalable IoT deployment
- Security monitoring in order to detect abnormal traffic volume and patterns
 - e.g. an unusual traffic spike for a normally quiet protocol
 - for both centralized or distributed architectures
- Network traffic monitoring and profiling
- Application traffic monitoring and profiling
- Capacity planning
- Security analysis
- Accounting
- Data warehousing and data mining

Challenges



Challenges with deploying flow analytics tools in an IoT network

- Flow analysis at the gateway **is not possible** with all IoT systems
 - LoRaWAN gateways simply forward MAC-layer sensor traffic to the centralized LoRaWAN network server, which means flow analysis (based on Layer 3) is not possible at this point.
 - A similar problem is encountered when using an MQTT server that sends data through an IoT broker
- Traffic flows are processed in places that **might not support** flow analytics, and visibility is thus lost.
- IPv4 and IPv6 native interfaces sometimes **need to inspect inside VPN** tunnels, which may impact the router's performance.
- Additional network **management traffic** is generated by analytics reporting devices

Thanks!

